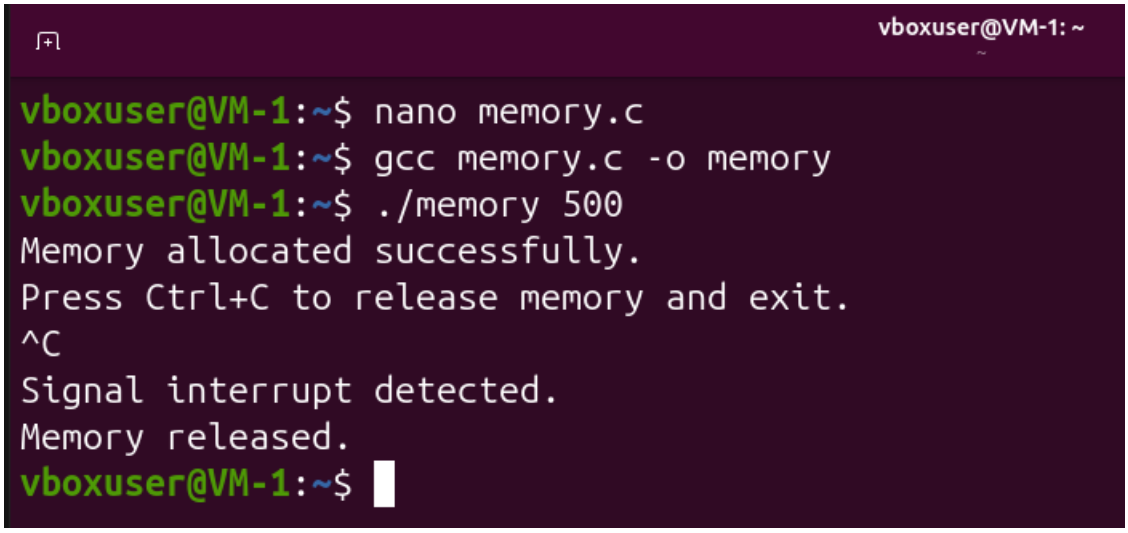
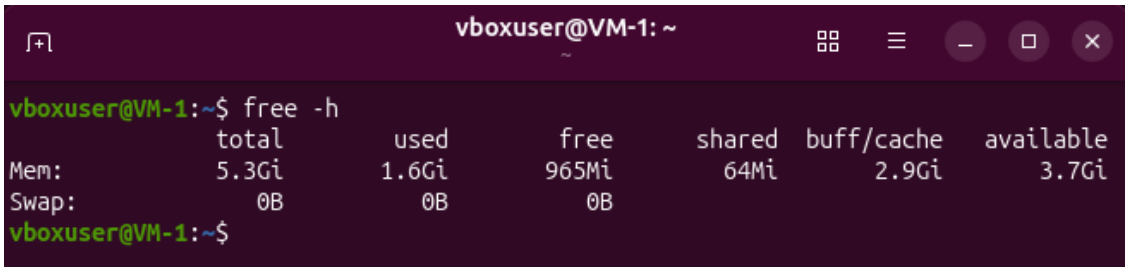
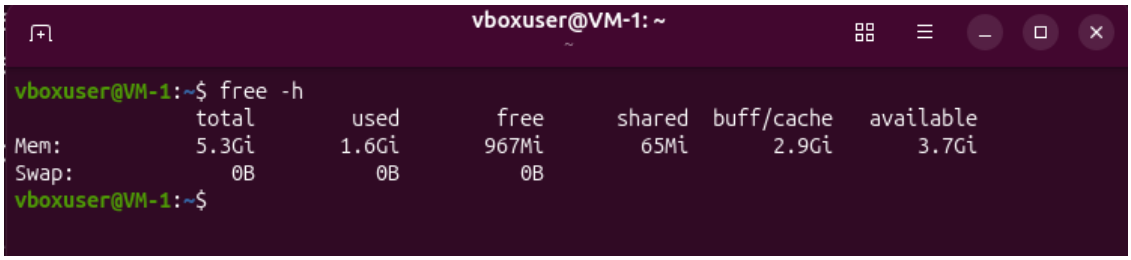
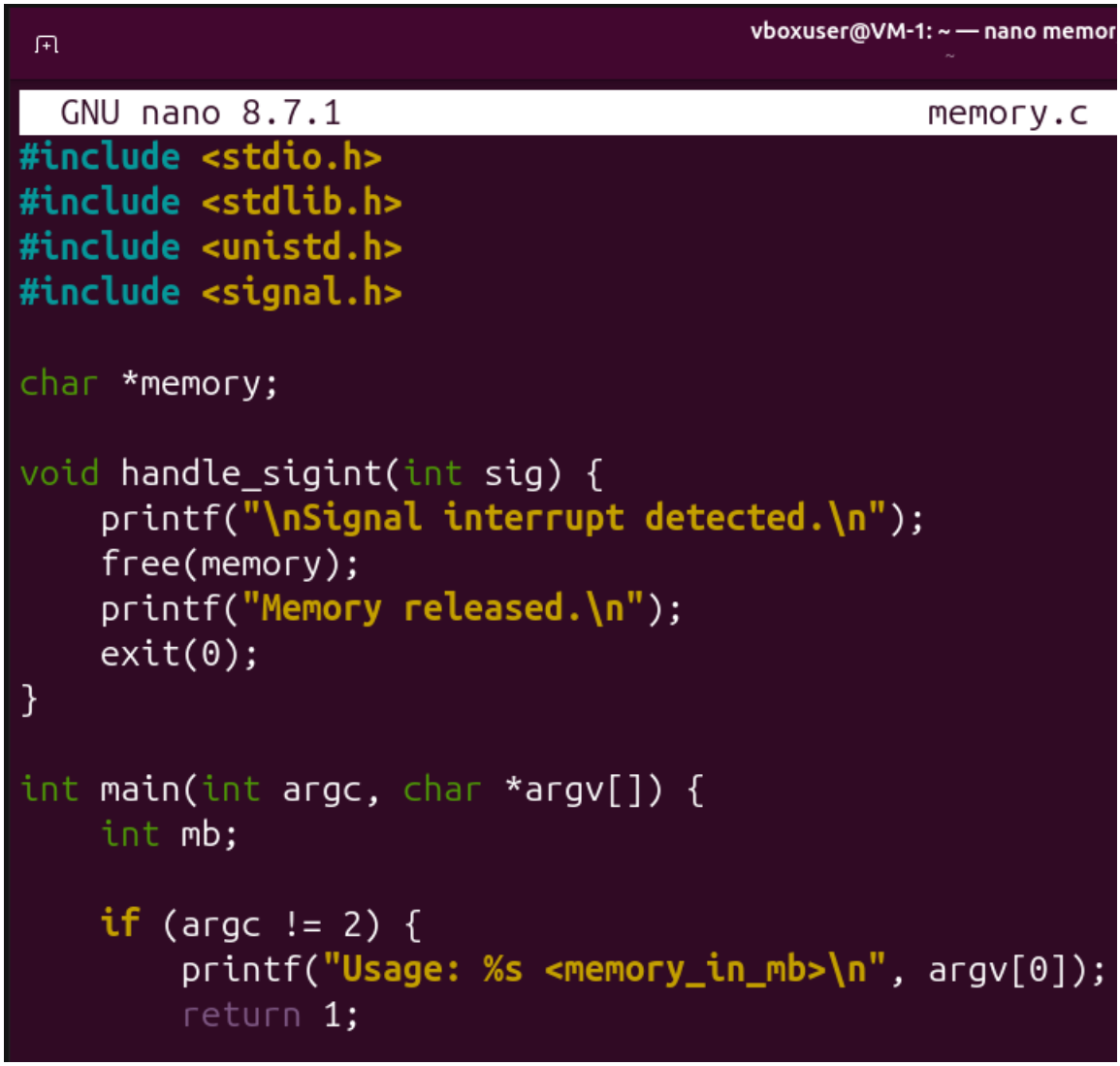
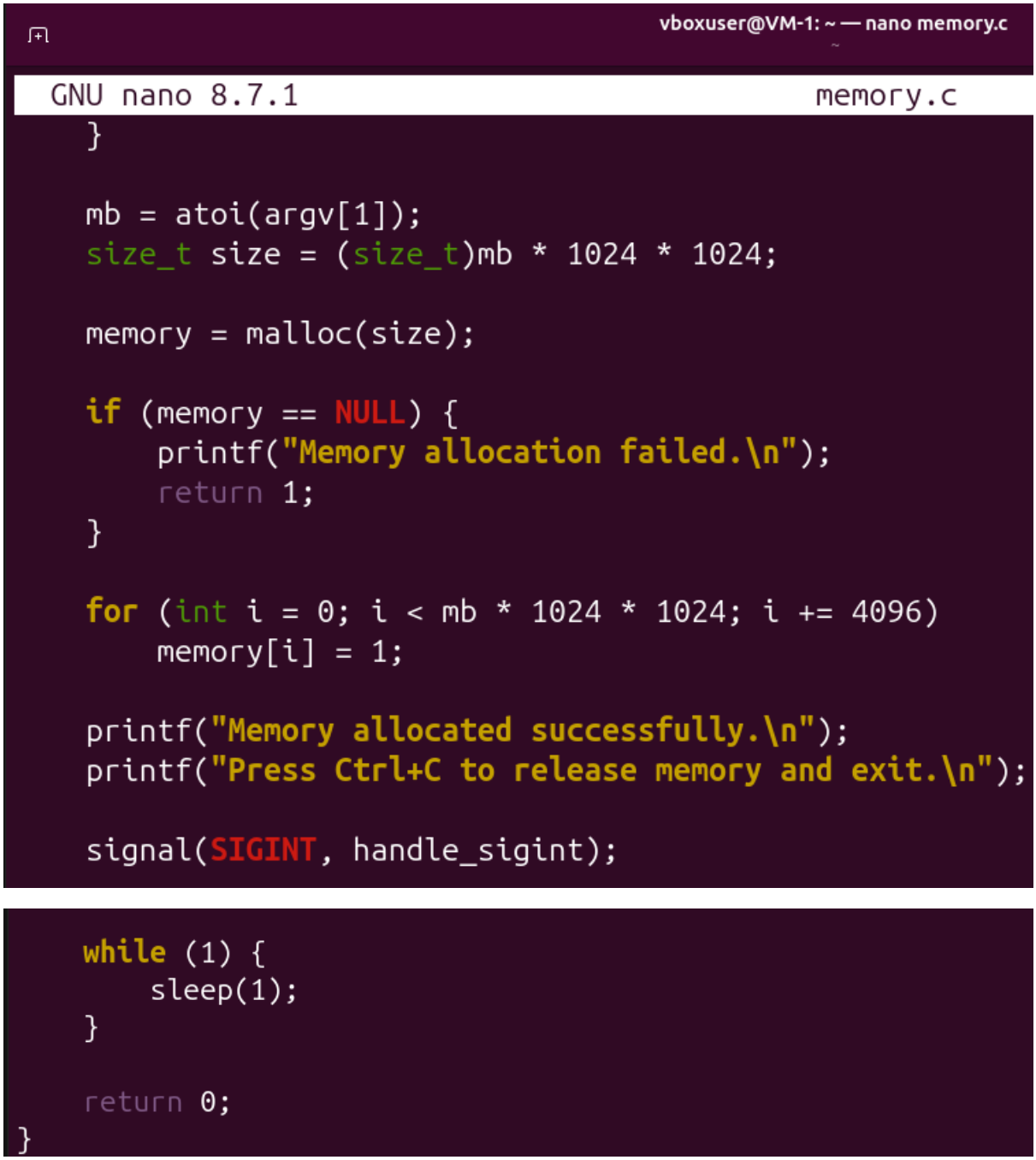

vboxuser@VM-1: ~
vboxuser@VM-1:~$ nano memory.c
vboxuser@VM-1:~$ gcc memory.c -o memory
vboxuser@VM-1:~$ ./memory 500
Memory allocated successfully.
Press Ctrl+C to release memory and exit.
^C
Signal interrupt detected.
Memory released.
vboxuser@VM-1:~$


vboxuser@VM-1: ~
vboxuser@VM-1:~$ free -h
total used free shared buff/cache available
Mem: 5.3Gi 2.1Gi 460Mi 66Mi 2.9Gi 3.2Gi
Swap: 0B 0B 0B
vboxuser@VM-1:~$


vboxuser@VM-1: ~
vboxuser@VM-1:~$ free -h
total used free shared buff/cache available
Mem: 5.3Gi 1.6Gi 965Mi 64Mi 2.9Gi 3.7Gi
Swap: 0B 0B 0B
vboxuser@VM-1:~$


vboxuser@VM-1: ~
vboxuser@VM-1:~$ free -h
total used free shared buff/cache available
Mem: 5.3Gi 1.6Gi 967Mi 65Mi 2.9Gi 3.7Gi
Swap: 0B 0B 0B
vboxuser@VM-1:~$


vboxuser@VM-1: ~ — nano memor
GNU nano 8.7.1 memory.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
char *memory;
void handle_sigint(int sig) {
printf("\nSignal interrupt detected.\n");
free(memory);
printf("Memory released.\n");
exit(0);
}
int main(int argc, char *argv[]) {
int mb;
if (argc != 2) {
printf("Usage: %s <memory_in_mb>\n", argv[0]);
return 1;




```
    while (1) {
        sleep(1);
    }

    return 0;
}
```